

An integrative multi-omics analysis of histone modifications and DNA methylation reveals the epigenomic landscape in apple under drought stress

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Summary

Epigenetic regulation plays a key role in plant development and stress response processes. Although previous studies have found that epigenetic modifications are involved in the drought response in apple, a comprehensive epigenomic profile of the response of apple to drought is needed. To characterize epigenomic profiles during the response of apple to drought, we conducted transcriptome, whole-genome bisulfite sequencing, and ChIP-seq analyses of six histone modifications (H3ac, H3K9ac, H3K14ac, H3K4me3, H3K27me3, and H3K36me3) in *Malus hupehensis* at 0, 3, 6, and 9 days after drought treatment. The greatest changes in differentially expressed genes were observed after 6 days of drought treatment. However, the highest levels of DNA methylation near the gene region were observed after only 3 days of drought treatment. The global enrichment of six histone modifications slightly decreased under drought treatment. Up-regulated drought-responsive genes with higher fold changes were associated with the hypo-regulation of H3K27me3, and up-regulated genes with lower fold changes were associated with the hyper-regulation of H3K4me3. Many drought-responsive genes such as *MYB88*, *NCED3*, and *JAZ1* are regulated by epigenetic modifications. We verified the functions of two candidate drought-responsive genes regulated by multiple epigenetic modifications, *MdABI5* (regulated by H3K14ac and H3K27me3) and *MdOCP3* (regulated by H3K9ac and H3K36me3) in the drought response. The phenotypes of transgenic apple under drought showed that *MdABI5* and *MdOCP3* positively regulate drought tolerance in apple. Our results provide new insights for studies of the molecular mechanism of epigenetic modifications and have implications for improving the drought resistance of apple.

Keywords: apple, drought, DNA methylation, histone modification, *MdABI5*, *MdOCP3*.

Introduction

The detrimental impact of abiotic stress on plants, instigated by instability and extreme weather events attributed to global warming, leads to over 38% of the global dryland continuing to expand (Huang *et al.*, 2016a). Drought becomes one of the most important abiotic stress factors which leads to huge crop loss. When plants are exposed to drought stress, they respond to drought through changes in physiological traits and molecular mechanisms. The molecular mechanisms of plant response to drought have been widely studied. There is an increasing trend of studies uncovering the function of genes and transcriptional regulations, as well as molecular regulatory networks involved in drought response (Geng *et al.*, 2018; Hou *et al.*, 2022; Shen *et al.*, 2022). Among them, an increasing number of studies have concerned epigenetic regulations, including DNA methylation, histone modification, and non-coding RNAs, which play an important role in the drought response process (Song

et al., 2024; Verslues *et al.*, 2023; Zhang *et al.*, 2022). As a favoured economic crop, apple is widely cultivated due to its high nutritional and economic value. Drought stress can significantly inhibit apple development, compromising both the fruit's quality and yield upon ripening. Understanding the molecular mechanisms of apples in response to drought stress is the basis for conducting the apple drought-resistance improvement (Li *et al.*, 2023c).

Epigenetics refers to changes in gene expression that are not caused by DNA sequence changes. Its regulation mechanism mainly relies on DNA methylation, modifications of histones, non-coding RNAs, and remodelling of chromatin. DNA methylation, as the fifth base of the genome, generally leads to gene silencing and functions in maintaining genome stability. Previous studies have shown that DNA methylation is also involved in plant responses to drought. For example, methylation levels in *Populus* were significantly increased near the gene region as well as in repeated sequences after drought treatment (Liang *et al.*, 2014).

Research on DNA methylation at different time points in strawberries has shown that DNA methylation may affect the expression of genes involved in ROS regeneration and scavenging during drought stress (Cao *et al.*, 2022). Single-base methylome analysis of drought-resistant and drought-sensitive apple cultivars under drought treatment reveals that dynamic DNA methylation changes in the apple genome in response to drought, and the changes in methylation levels of abundant genes and transposable elements (TEs) were associated with drought (Xu *et al.*, 2018). A recent study demonstrated that methylation on MITE insertion in the *MdRFNR1-1* promoter could activate its expression by recruiting the MdSUVH1-MdDNAj complex and positively regulate drought resistance in apple (Niu *et al.*, 2022). These results showed that DNA methylation modification on genes or TEs could regulate their expression to participate in the drought response.

In addition to DNA methylation, histone modification is a covalent post-translational modification of histone proteins, including acetylation, methylation, phosphorylation, and so on. Several studies have shown that various histone modifications are involved in the plant response to drought (Ueda and Seki, 2020). For example, studies have shown that H3K9ac levels increase during drought and decrease rapidly upon rehydration. In contrast, histone H3K4me3 gradually decreases during drought and remains at low levels even after rehydration. This suggests that H3K4me3 serves as an epigenetic marker for drought stress (Kim *et al.*, 2012; van Dijk *et al.*, 2010). Besides acting individually, there is also the interplay between histone modifications (Ueda and Seki, 2020). Many genes show significant histone marks switching during their response to drought (Widiez *et al.*, 2014). H3K4me3 and H3K27me3 markers have the role of a molecular switch but function independently and are not mutually exclusive on the genes responsive to drought stress (Liu *et al.*, 2014; Widiez *et al.*, 2014). The abundance of H3K4me3 and H3K9ac upstream of drought-responsive genes *RD29A*, *RD29B*, *RD20*, and *RAP2.4* in *Arabidopsis* correlates with their activation during drought stress (Kim *et al.*, 2008). It is widely known that the histone modification process is controlled by histone modification catalytic enzymes, called 'histone modifiers'. Generally, the histone modifiers participate in the drought response process by inducing the histone modification changes in drought-responsive genes. *Arabidopsis* histone methyltransferase ATX1 can bind to the drought-responsive genes *NCED3*, *RD29A*, *COR15A*, and *ADH1* and enhance the enrichment of H3K4me3 levels, resulting in the gene activation of these genes under drought (Ding *et al.*, 2011). A recent study has uncovered that an apple histone deacetylase MdHDA6 can facilitate histone deacetylation on drought-responsive genes including *ABI5*, *YUC6*, *RD29A*, *DREB2A*, and so on to inhibit their expression and negatively regulate drought resistance in apple (Li *et al.*, 2023a). These results reveal that histone modifications controlled by various histone modifiers can affect the expression of drought-related genes involved in drought response.

The plant hormone abscisic acid (ABA), as one of the core hormones in drought response, has an important role in regulating processes such as stomatal closure, balanced growth, and stress tolerance (Zhang *et al.*, 2022). ABA INSENSITIVE 5 (*ABI5*) is a basic leucine zipper (bZIP) transcription factor reported to be involved in plant development and abiotic stress responses (Skubacz *et al.*, 2016). *ABI5* has been reported to improve water use efficiency (WUE) and positively regulate drought tolerance in

plants (Liu *et al.*, 2022; Mittal *et al.*, 2014). *MdABI5* is repressed by the transcription factor *MdTCP46* to regulate drought tolerance in apples (Liu *et al.*, 2022) and regulate histone acetylation modification upstream of drought-regulated genes by interacting with *MdHDA6* (Li *et al.*, 2023b; Wang *et al.*, 2023). OVEREXPRESSION OF CATIONIC PEROXIDASE 3 (*OCP3*) is a homeodomain (HD) transcription factor reported to be involved in stress response in *Arabidopsis* (García-Andrade *et al.*, 2011; Ramírez *et al.*, 2009). The *ocp3* mutant enhanced plant resistance through ABA and JA in *Arabidopsis thaliana* (García-Andrade *et al.*, 2011; Ramírez *et al.*, 2009). The *ocp3* mutant promotes stomatal closure through the ABA signalling pathway and is more drought resistant (García-Andrade *et al.*, 2011). However, there have been no studies on the regulation of *MdABI5* and *MdOCP3* by histone modification.

The prospective application of epigenetic variations in crop improvement has been gradually concerned and attached importance (Lloyd and Lister, 2022; Qi *et al.*, 2023). The first step of utilizing epigenetic variation in apple drought-resistance improvement is to uncover the mechanisms of epigenetic regulations involved in drought response. In a previous study, we revealed the DNA methylation changes during drought treatment in apple (Xu *et al.*, 2018), but the dynamics of various epigenetic modifications during apple in response to drought are still unknown. In this study, we applied WGBS (whole-genome bisulfite sequencing), ChIP-seq (chromatin immunoprecipitation sequencing), and ssRNA-seq (strand-specific sequencing of RNA) to investigate the involvements of DNA methylation, histone modification, as well as non-coding RNAs in *Malus hupehensis* seedlings under different drought treatment periods. We first analysed the global changes of DNA methylation, histone modifications, and gene expression during drought treatment, and then investigated the epigenetic variations on the drought-responsive genes. Moreover, we also explored the correlation among different epigenetic modifications during the drought response process. Our results revealed the potential epigenetic regulation mechanisms of apple's response to drought, providing a base for apple drought-resistance improvement.

Results

Dynamic gene expression changes under different drought treatments in apple

To reveal the contributions of changes in epigenetic modifications in apple under drought treatment, 3-month-old *Malus hupehensis* seedlings were subjected to a drought treatment experiment. WGBS, ChIP-seq analyses of six histone modifications (H3ac, H3K9ac, H3K14ac, H3K4me3, H3K27me3, and H3K36me3), ssRNA-seq, and RNA-seq were performed on *Malus hupehensis* after 0 days (A0d), 3 days (A3d), 6 days (A6d), and 9 days (A9d) of drought treatment (Figure 1a). Transcriptome analysis revealed 3041 differentially expressed genes (DEGs) at A3d compared with the control A0d (A3d vs. A0d), suggesting that several genes had responded to drought stress in the early stages of drought treatment. A total of 3818 and 3816 DEGs were identified at A6d (A6d vs. A0d) and A9d (A9d vs. A0d) (Figure 1b). Moreover, the number of down-regulated genes was higher than the number of up-regulated genes in the early drought response (A0d–A6d), and the number of up-regulated and down-regulated genes was similar in the response to A9d (Figure 1b). To explore changes in

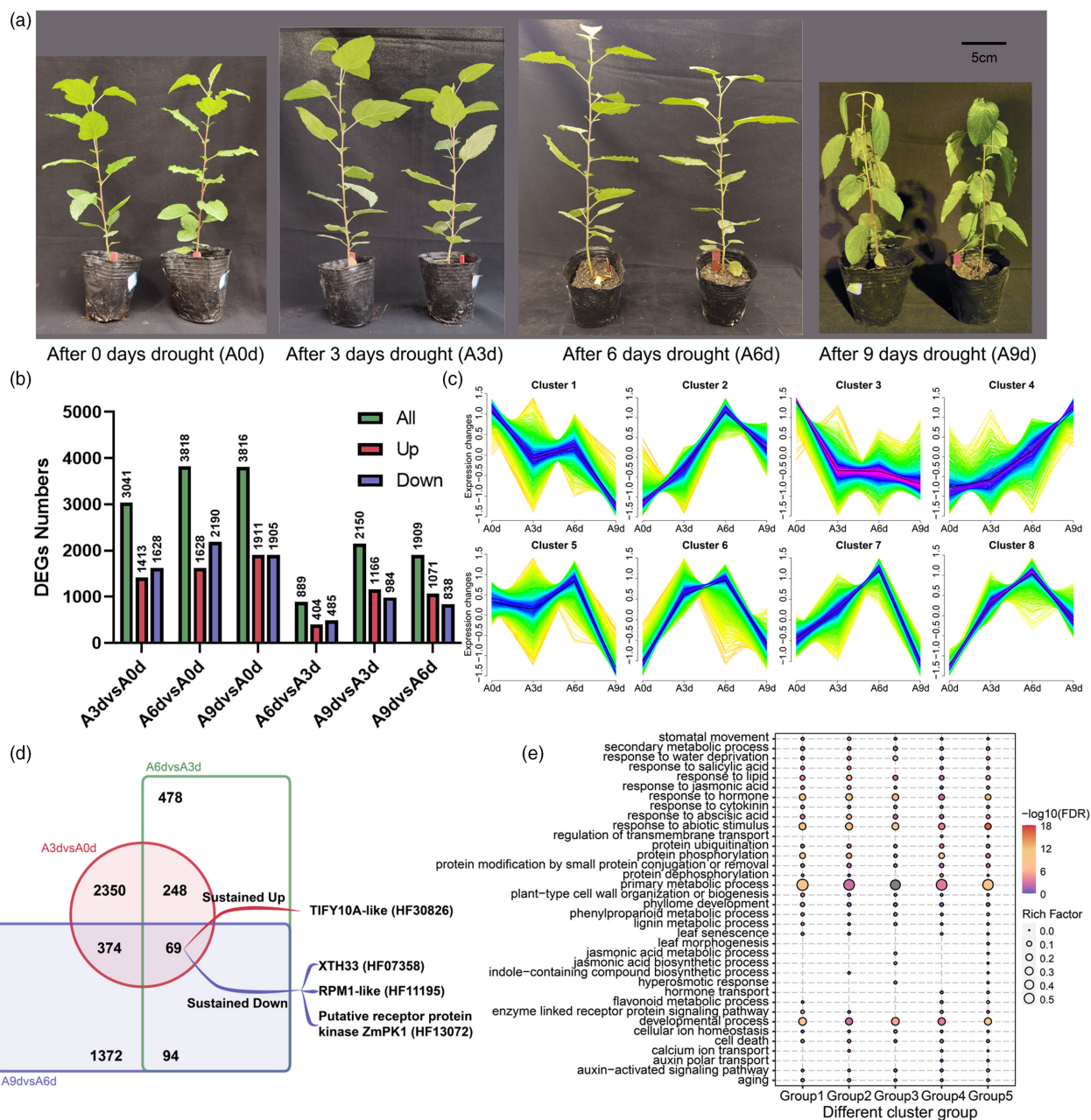


Figure 1 Transcriptome differential analysis, clustering, and functional enrichment results. (a) The phenotype of drought treatment for 0, 3, 6, and 9 days in *Malus hupehensis*. (b) Numbers of DEGs in different periods of differential comparison combinations. (c) Clustering of expression trends of DEGs in different periods. (d) Overlap of DEGs in adjacent comparison combinations under drought treatment. (e) Functional enrichment comparison dot plot for different expression trend gene groups (Group 1: Cluster 1, Group 2: Cluster 3, Group 3: Cluster 4, Group 4: Cluster 5, Group 5: Clusters 2, 6, 7, and 8).

gene expression levels during different drought-treatment periods in apple, we clustered all DEGs based on their expression trends in response to drought. The clustering results showed that 56.5% of the genes were up-regulated. The expression of most genes was down-regulated in A9d compared with A6d (Figure 1c). To identify genes that play a crucial role in the drought response, we characterized overlap among all the DEGs that showed consistent responses to drought treatment. A total of 69 genes with consistently differential expression during drought treatment were identified, and the expression of four genes (*TIFY10A-like*, *XTH33*, *RPM1-like*, and *PK1*) was consistently up-regulated or

down-regulated (Figure 1d). These four genes have been reported to potentially participate in the drought response (Bassett et al., 2005; De Caroli et al., 2021; Haider et al., 2017).

Genes with different drought response trends may have different functions during drought; we thus clustered the genes with different trends into five groups (Group 1: Cluster 1, Group 2: Cluster 3, Group 3: Cluster 4, Group 4: Cluster 5, Group 5: Clusters 2, 6, 7, and 8) and performed functional enrichment analysis. Each group was significantly enriched in pathways related to water deprivation, but the water deprivation pathway was most significantly enriched in Group 3 (Figure 1e). In addition

to being significantly associated with water deprivation, genes in each of these groups were also specifically associated with different biological pathways. Genes in Group 1 were significantly enriched in pathways related to leaf senescence, leaf development, salicylic acid response, and cell wall synthesis. Genes in Group 2 were significantly enriched in the aforementioned pathways, as well as various hormone-related pathways and stomatal movement pathways. Genes in Group 3, which were consistently up-regulated, showed higher enrichment in ion homeostasis and the jasmonic acid biosynthesis pathway. No expression changes were observed in Group 4 genes during A0d–A3d; the expression of these genes was rapidly up-regulated in A6d, and the expression of these genes declined in A9d. These genes were notably enriched in protein modification processes. Group 5 genes were most significantly enriched in abiotic stress responses; they were also significantly enriched in pathways such as hormone transport, primary metabolism, development, and aging (Figure 1e; Table S1). The clustering results showed that the expression levels of most genes peaked in A6d, and these genes were closely related to pathways such as water deprivation. This may indicate that A6d is a key stage in the drought response in apple.

Long non-coding RNA (lncRNA) is more than 200 nucleotides in length and lacks protein-coding potential. lncRNAs have been reported to play a role in stress responses in plants (Di et al., 2014; Jha et al., 2020). Based on the ssRNA-seq data, we identified lncRNAs and their expression changes in apple under drought treatment (Figure S1). The results showed that lncRNAs have fewer exons, are shorter, and lack open reading frames (ORFs) compared with mRNAs (Figure S1A). Previous studies have shown that lncRNAs are important endogenous targets, which reduce the targeting effect of miRNAs on mRNAs (Zhu and Wang, 2012). Based on the differentially expressed miRNA data, we characterized the lncRNA-miRNA-mRNA targeting relationship in the A6d versus A0d comparison group. The results showed that these mRNAs included drought-responsive genes, such as *TPL* (Ke et al., 2015), *ANAC087* (Huysmans et al., 2018), and the protein kinase superfamily protein gene. This indicates that these genes might be regulated by the antagonistic action of lncRNAs and miRNAs (Figure S1B). Studies on lncRNAs have also reported that they play a role in *cis*-regulation and *trans*-regulation and are involved in various biological processes (Heo et al., 2013). In our study, we identified both the *cis*-regulated targets and *trans*-regulated targets of lncRNAs. These genes are preferentially enriched in response to stress, metabolic processes, and protein phosphorylation (Table S2). We also determined the overlap among DEGs, lncRNA targets, and miRNA targets to identify their potential regulatory roles, and 57.8% of DEGs may be regulated by lncRNAs and miRNAs (Figure S1C). Eight DEGs were targeted by lncRNAs and miRNAs, and most of these genes were reported to be involved in stress responses, including CYP98A3 (Schoch et al., 2001), ZAT12 (Rai et al., 2012), and WRKY31 (Jang et al., 2010) (Table S3). To further clarify the central role of lncRNAs, we constructed expression regulatory networks based on lncRNA and mRNA expression correlations. In the A3d versus A0d comparison group, five lncRNAs (TCONS_00019097, TCONS_00168895, TCONS_00027359, TCONS_00175556, and TCONS_00021790) may play important roles in plant drought responses, which are associated with multiple WRKY family transcription factors and metabolism-related genes (Figure S1D).

Similarly, in the other comparison groups, related lncRNAs were associated with stress and metabolism-related genes (Table S4), suggesting that lncRNAs may play an important role in the apple drought response.

Landscape of DNA methylation in the response of apple to drought stress

Previous studies have revealed that DNA methylation plays an important role in the plant drought response (Li et al., 2023c; Xu et al., 2018). To investigate the global changes in DNA methylation in apple under drought, we performed DNA methylome analysis at the single-base resolution to reveal DNA methylation changes in response to drought (Figure 2). Generally, significant increases were observed in the global DNA methylation levels of mCG, mCHG, and mCHH in all three stages (A3d, A6d, and A9d) of the drought treatment compared with the control (A0d) (Figure 2a). At a genome-wide scale, DNA methylation levels near the gene regions increased at 3 days of drought; this was followed by a decrease at 6 days of drought and then a slight increase after 9 days of drought (Figure 2a). The DNA methylation level changed significantly at the beginning of the drought treatment (Figure 2a), which indicates that DNA methylation plays a key role in the early stage of the apple drought response. The number of different methylation regions (DMRs) changed drastically at A3d and then gradually decreased at A6d and A9d, suggesting that the effect of DNA methylation decreased during drought stress (Figure 2b). Near the gene regions, DNA methylation generally affects the expression of genes by affecting the DNA methylation level in promoter regions. Analysis of the transcriptome data indicated that DMRs and DEGs showed opposite patterns of change in the A3d versus A0d, A6d versus A0d, and A9d versus A3d comparison groups, which is consistent with the inhibitory effect of DNA methylation in the promoter regions (Figure 2b). To explore the contribution of DNA methylation in the response of apple to drought stress, we conducted enrichment analysis of genes associated with DMRs within the 2000-bp upstream regions. The results showed that the CHG-related genes of A3d versus A0d and A6d versus A3d, as well as the CHH-related genes of A6d versus A0d, were significantly up-regulated in response to water deprivation. We also observed that hypo-regulated DMRs in the A6d versus A0d comparison group and hyper-regulated DMRs in the A9d versus A0d comparison group are significantly enriched in response to water deprivation (Table S5). The enrichment results revealed that DNA methylation plays an important role in apple drought response. The regulated genes included drought response genes such as *ABI4*, *CIPK6*, and *SOS3*. In addition to the response to water deprivation, DMR-related genes were enriched in metabolic processes, plant growth and development, and various hormone responses (Table S5). Our data suggested that changes in DNA methylation under drought treatment play a critical role in the response of apple to drought.

We also analysed the DNA methylation distribution in the whole genome under drought stress. There were no significant changes in the distribution of DMRs at the genomic level for mCpG and mCHG. However, there was an increase in mCHH within the 2000-bp upstream region of genes in the A6d versus A3d comparison group (Figure 3c). Based on the results of the DMR-related gene enrichment analysis (Table S5) and DNA methylation level changes (Figure 2a), mCHH appears to play a role in the drought stress response. mCHH is mainly established

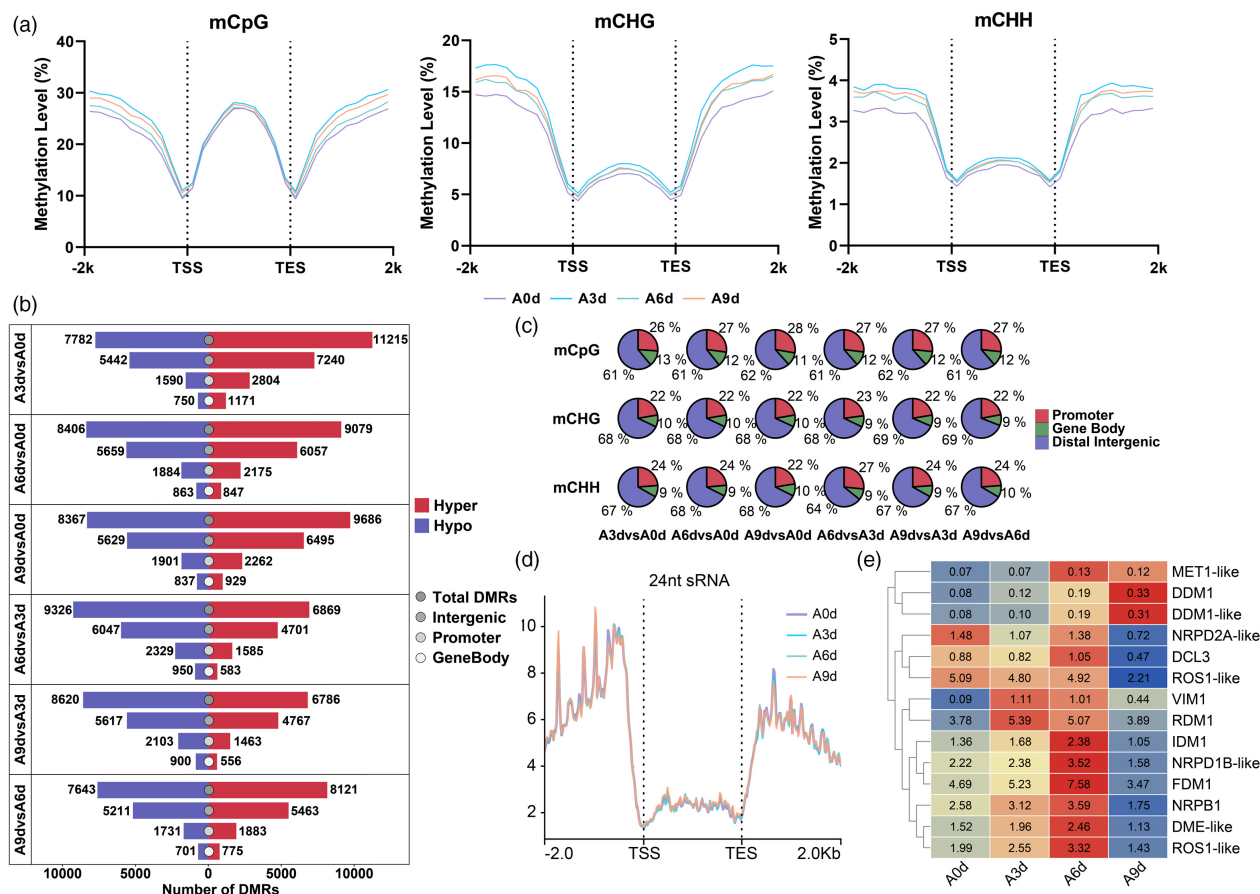


Figure 2 Characterization of WGBS, 24 nt siRNA distribution, and related methyltransferase changes in response to drought. (a) DNA methylation patterns near the gene body at whole genome level. (b) Number of hyper- and hypo-regulated DMRs related genes under different drought comparison combinations. (c) The percentage of each methylation type distribution for different comparative combinations at the whole genome level. (d) Enrichment of 24 nt siRNA near gene body regions. (e) Expression patterns of differentially expressed DNA methylation-related factors under drought treatment.

through RNA-directed DNA methylation (RdDM) and is highly associated with 24-nt siRNAs. To investigate whether the increase in mCHH is caused by the hyper-regulation of the RdDM pathway, we characterized changes in the enrichment of 24-nt siRNAs. No significant changes in the 24-nt siRNAs near the gene region were observed, except in a few regions at the genomic level (Figure 2d). DNA methylation is known to be controlled by a series of DNA methyltransferases/demethylases. To further explore the cause of drought-induced DNA methylation changes, we investigated gene expression changes involved in the DNA methylation pathway, including *VIM1*, *MET1-like*, *NRPB1* associated with CpG methylation, *RDM1* related to RNA polymerase, *DDM1* involved in maintaining symmetric methylation, *DCL3* associated with the RdDM pathway, and *ROS1-like*, *DME* and *IDM1* enzymes involved in DNA demethylation (Zhang et al., 2018). Each of these genes only partially contributes to changes in the DNA methylation level under different drought treatments (Figure 2a,e). The increase in the DNA methylation level at A0d to A3d may be related to the increased expression of genes involved in maintaining methylation and RNA polymerase, and the decreased expression of genes at A3d to A6d may be partially influenced by DNA demethylases such as DME1-like, which control changes in DNA methylation during

drought processes. The DNA methylation level changes at A6d to A9d may be related to DDM1 and DNA demethylase (Figure 2).

Profiles of histone modification variations involved in the response of apple to drought

To further reveal the regulatory role of various histone modifications in the apple drought response, we performed ChIP-seq of six histone modifications, including H3ac, H3K9ac, H3K14ac, H3K4me3, H3K27me3, and H3K36me3, after 0, 3, 6, and 9 days of drought treatment. All six types of histone modifications were highest after TSS in the 2 k region near the gene region. Modifications such as H3K4me3, H3K9ac, and H3K14ac were slightly lower in A6d compared with A0d and A3d (Figure 3a; Figure S2A). These histone modifications were primarily distributed in gene regions and promoter regions, which are consistent with their distribution characteristics (Figure S2B).

We identified the DEGs associated with the histone modification variations to explore the potential role of histone modification in the drought response. The DEGs involved in drought response processes contained histone acetylation/deacetylation and methylation/demethylation modifications. The heatmap shows that most of the histone-related

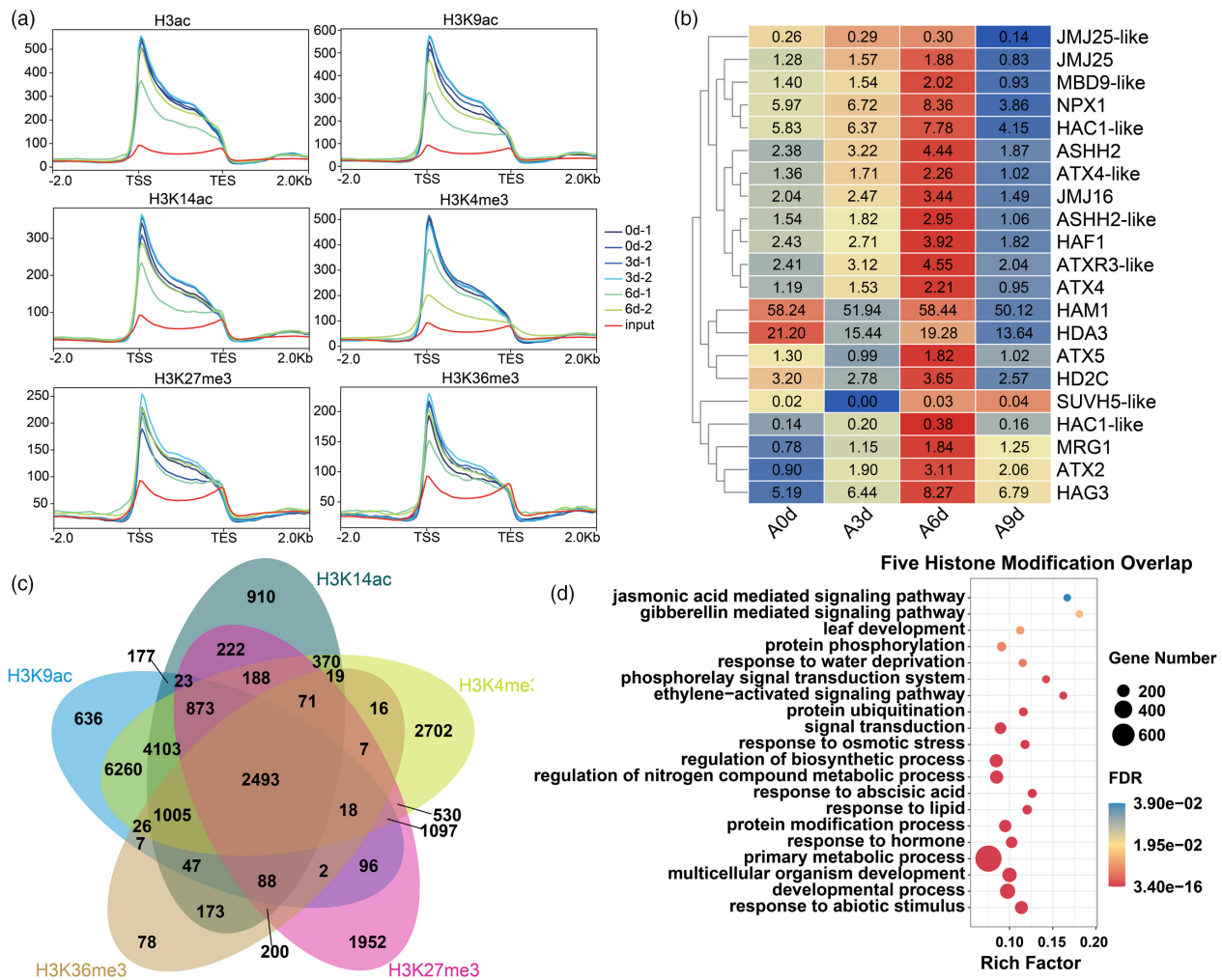


Figure 3 Characterization and differential analysis of histone modifications. (a) Distribution of various histone modifications around the gene region. (b) Expression patterns of differentially expressed histone modification-related factors under drought treatment. (c) Overlap of drought-responsive genes regulated by histone modifications. (d) Functional analysis of histone modification enrichment preference genes under drought treatment.

genes had higher expression levels in A6d (Figure 3b). This suggests that the slight decrease in histone modification at A6d may be influenced by various factors. Next, we identified DEGs related to differentially histone-modified regions (DHMRs). The 2493 genes showing overlap may be co-regulated by five types of histone modifications under drought treatment, with each histone modification accounting for 12.6%–57.7%. The largest proportion of genes independently regulated by a single histone modification was 24.6% (for H3K27me3), which may indicate that the correlations between the activation markers (H3K9ac, H3K14ac, H3K4me3, and H3K36me3) may be stronger than that of the inhibitory marker H3K27me3 (Figure 3c). We performed GO enrichment analysis of 2493 genes, which were shown to be associated with water deprivation, response to hormones, protein phosphorylation, ubiquitination, and metabolism (Figure 3d; Table S6). Among the 2493 genes, only 28.8% of them are DEGs. Even after removing the DEGs, the remaining genes were significantly enriched in water deprivation (Table S6). The results suggest that histone modification regulation may respond to drought not only by altering transcript abundance but also through other mechanisms in the drought response.

Contribution of histone modifications and DNA methylation changes to gene expression variation in the response to drought

Analysis of DEGs indicated that there was a significant change in gene expression in A6d compared with A0d under drought treatment. The results of the analysis of DNA methylation and six histone modifications revealed that significant changes in DNA methylation occurred at A6d compared with A0d, and potential correlations were observed among histone modifications. Based on the high-throughput sequencing data, we first investigated the distribution of the DNA methylation level and six histone modifications (Figure S2C,D). The results revealed high overlap in the overall distribution of histone modifications with the distribution of genes. The distribution of H3K9ac and H3K14ac was similar to the overall histone acetylation level (H3ac). However, there were differences in the distribution of the three types of histone methylation in some regions. Opposite patterns in H3K4me3, H3K36me3, and H3K27me3 were observed in some regions, indicating that they may have different regulatory mechanisms (Figure S2C). The DNA methylation distribution across the

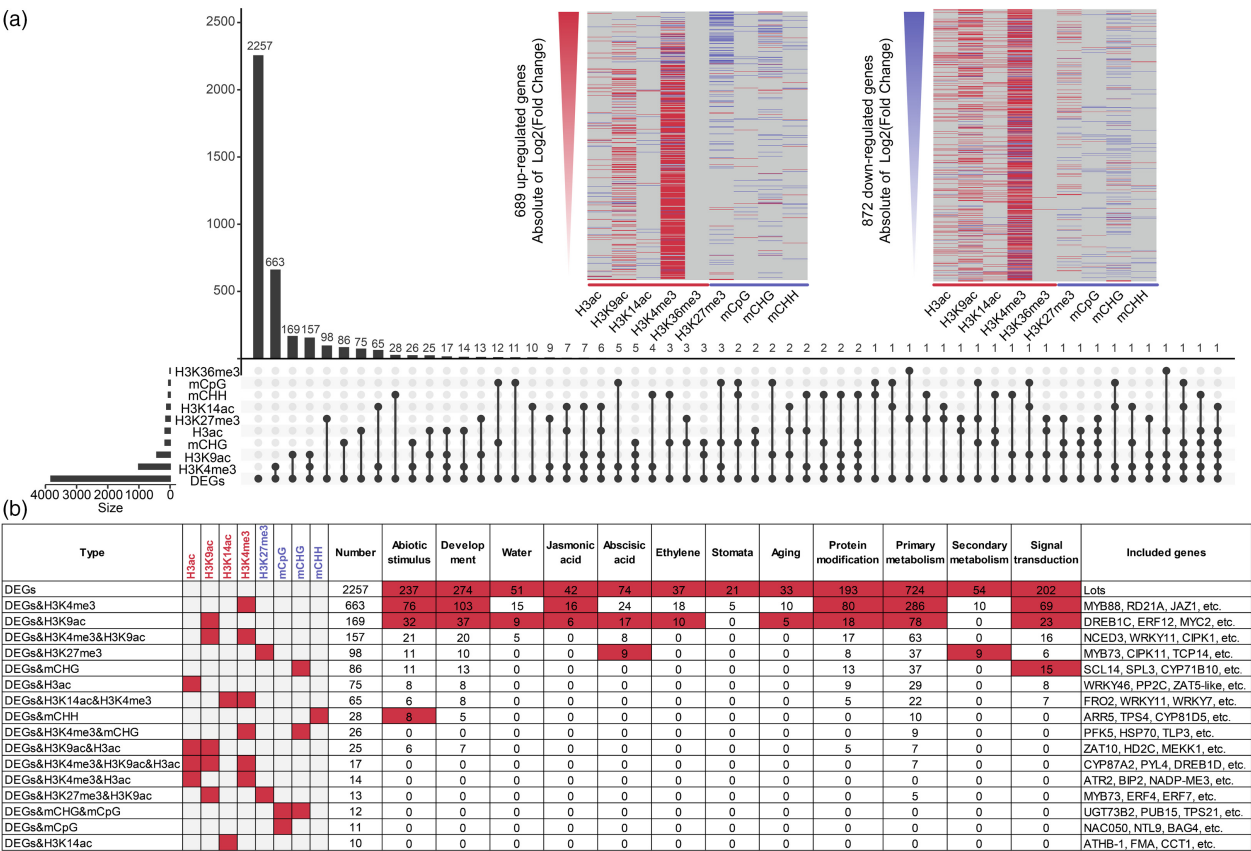


Figure 4 Relationship between six histone modifications, DNA methylation, and their regulatory functions under drought treatment. (a) Upset plot of differential expression genes co-regulated by DNA methylation and histone modifications as well as the expression changes of the regulated genes. Heatmap of the relationship between all up- and down-regulated DEGs regulated by an epigenetic modification, the log₂ (Fold Change), and histone modification. Red represents a positive correlation with epigenetic modification, while blue represents a negative correlation. (b) Functional enrichment tables for genes regulated by different epigenetic modifications. In the function enrichment part, the red box represents a significant enrichment in the GO pathway, while white represents no significant.

genome was opposite to that of the six histone modifications and the gene regions (Figure S2D). There were no obvious synergistic correlations between DNA methylation and the six histone modifications at the genome-wide level, which is related to their distributional characteristics in the genome. However, correlations between DNA methylation and six types of histone modifications may be observed in the transcriptional regulation of some genes.

To explore correlations between DNA methylation and histone modifications under drought treatment, we analysed the proportions of different epigenetic modifications after drought treatment. Analysis of different epigenetic modification features (DMRs or DHMRs) combined with DEGs revealed that 36.6% of DEGs were regulated by DNA methylation and six types of histone modifications, and 17.1% and 19.5% of these DEGs were up-regulated and down-regulated in A3d versus A0d, respectively. During drought stress, the proportion of DEGs regulated by histone modifications was highest in A6d versus A0d, reaching 40.9%. Significant differences in many epigenetic regulators were observed in A6d versus A0d, including 39.1% of genes in A6d versus A0d regulated by DNA methylation and 40.9% of genes in A6d versus A0d regulated by histone modifications. A6d was one of the most important stages in the response to drought at the transcriptional level. Therefore, we focused on the A6d

versus A0d comparison under drought treatment. Upset plots showed that the number of genes regulated by H3K4me3 was significantly higher than that of the other epigenetic markers under drought treatment (Figure 4a). Targeted genes are mostly regulated by differentially enriched histone modifications and DNA methylation. Among DEGs involved in the response to drought, 29.9% of these DEGs were regulated by a single epigenetic mark, 9.2% of these DEGs were regulated by two epigenetic marks, and less than 2% of these DEGs were regulated by multiple epigenetic modifications. Approximately 2.5% of DEGs were co-regulated by both activating and repressing epigenetic marks. In the genes regulated by DNA methylation and histone modifications, H3K4me3 and mCHG were more likely to co-regulate genes. Between the six-histone modification regulated genes, activating histone modifications, including H3K4me3 and H3K9ac, likely co-regulate genes. Conversely, co-regulation by both activating and repressive histone modifications, such as H3K27me3, H3K4me3, and H3K9ac, is relatively uncommon (Figure 4a). For all the DEGs regulated by epigenetic modifications, a heatmap is presented to show their overall changes, which are consistent with the regulatory characteristics of various epigenetic modifications. Up-regulated genes with lower fold changes were more likely regulated by H3K4me3, whereas genes with higher fold changes were more likely

regulated by the hypo-regulation of H3K27me3. However, this pattern was not observed in the down-regulated genes (Figure 4a).

To explore associations between different histone modifications and DNA methylation, we conducted a functional analysis of different groups varying in histone modifications and DNA methylation levels. Our epigenetic modification data can only explain 40.9% of the changes in DEGs, but the remaining DEGs were also significantly enriched in various pathways related to drought (Figure 4b). DEGs regulated by H3K4me3 were significantly enriched in jasmonic acid and protein modification pathways, such as *MYB88*, which regulates cell wall and hydraulic conductivity (Geng *et al.*, 2018), and *JAZ1*, which is involved in drought responses through the JA and ABA pathways (Fu *et al.*, 2017). DEGs regulated by H3K9ac play important roles in the response to jasmonic acid, ABA, ethylene, and water deprivation, such as *DREB1C* and *MYC2*, which are reported to be involved in drought tolerance (Abe *et al.*, 2003; Novillo *et al.*, 2004). H3K27me3 only regulates a small number of DEGs, but it is significantly enriched in both the response to ABA and secondary metabolic processes, which indicates that H3K27me3 plays an important role in drought response processes (Figure 4b). Many drought-responsive genes were regulated by epigenetic modifications, indicating that epigenetic modifications play an important role in drought responses.

Histone modification-regulated MdABI5 positively regulates drought tolerance in apple

Previous studies have shown that ABI5 plays an important role in plant drought tolerance by targeting PYLs to regulate the ABA response pathway (Lopez-Molina *et al.*, 2001; Zhao *et al.*, 2020). However, the role of *MdABI5* in drought resistance and the epigenetic regulatory mechanism of *MdABI5* have not been studied. *MdABI5* is positively regulated after drought stress (Figure 5a), and this is associated with an increase in the enrichment level of upstream H3K14ac and a decrease in H3K27me3 (Figure 5b). To further validate the ChIP-seq results, we performed ChIP-qPCR validation of histone modifications upstream of ABI5. The results were consistent with the ChIP-seq data (Figure 5c). We generated *MdABI5* overexpression (*MdABI5*-OE) transgenic lines (Figure 5d) in apple and performed drought treatments on transgenic lines and GL-3 (wild-type) to observe the function of *MdABI5* in whole plants during drought stress. The *MdABI5* expression levels of *MdABI5*-OE lines were 35 to 50 times higher than those of wild-type plants (GL-3) (Figure 5e). Under short-term drought treatment, the phenotypic results showed that the *MdABI5*-OE lines exhibited a more drought-resistant phenotype compared with wild-type plants (Figure 5f).

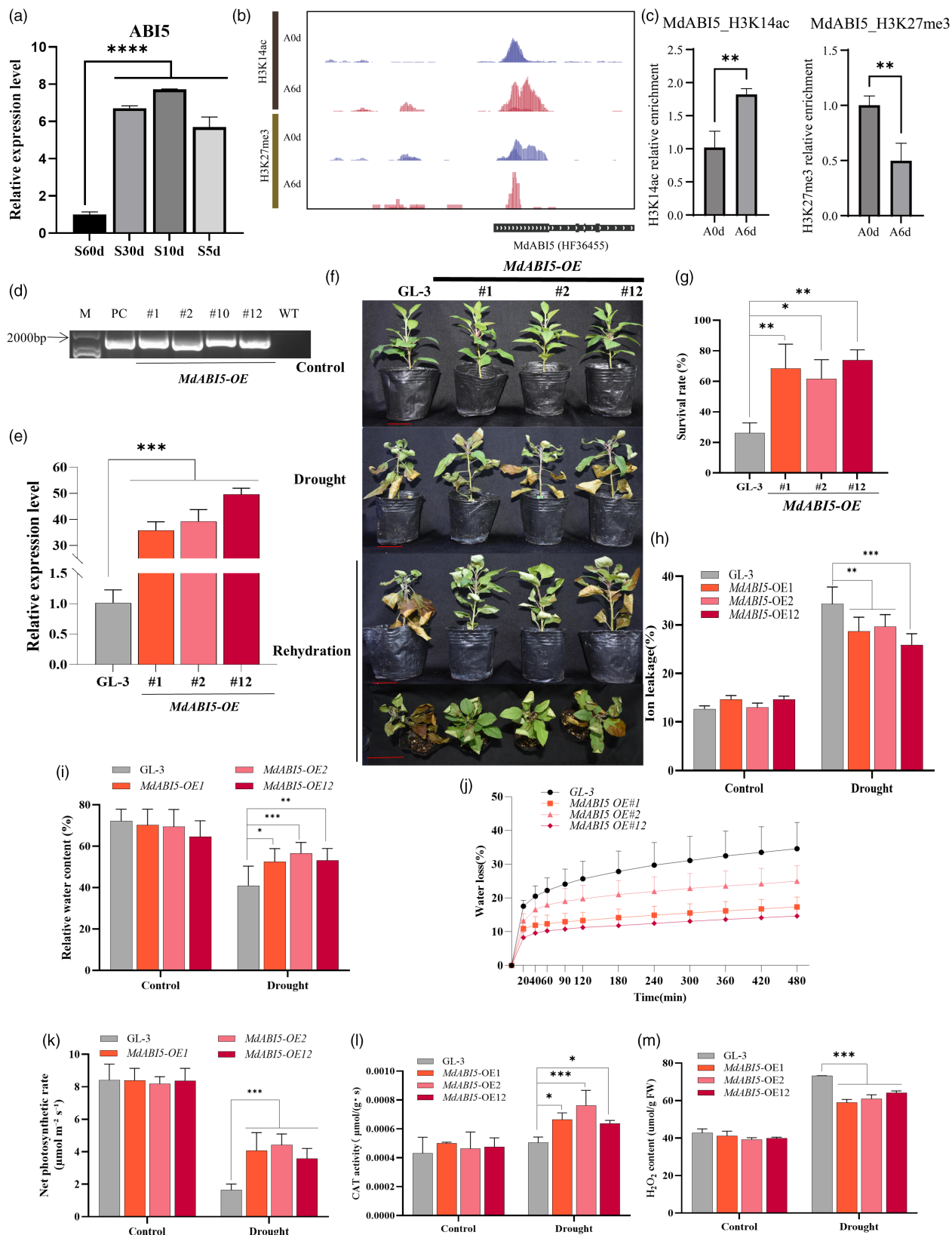
After 7 days of rehydration, the survival rate of *MdABI5*-OE lines was significantly higher than that of wild-type plants (Figure 5g). After drought stress, *MdABI5*-OE plants showed significantly lower ion leakage than wild-type plants (Figure 5h), indicating that the cell membranes of *MdABI5*-OE plants were less damaged than those of wild-type plants. The relative leaf water content (RWC) after drought stress was significantly higher than that of wild-type plants, and no differences were observed under normal conditions (Figure 5i). Analysis of the water loss of detached leaves revealed that the rate of water loss of *MdABI5*-OE was lower than that of wild-type plants, indicating that *MdABI5*-OE leaves lost water more slowly (Figure 5j). The *MdABI5*-OE lines had higher net photosynthetic rates (Figure 5k), higher catalase (CAT) activity (Figure 5l), and lower hydrogen peroxide (H_2O_2) accumulation under drought stress compared with wild-type plants (Figure 5m), suggesting that the *MdABI5*-OE lines were less affected by drought stress compared with wild-type plants. These results indicated that up-regulation of *MdABI5* reduced leaf water loss and mitigated damage under drought treatment compared with wild-type plants.

To comprehensively elucidate the function of *MdABI5* under drought treatment, we generated *MdABI5* RNA-interfering (*MdABI5*-RNAi) transgenic lines (Figure 6a,b) and analysed the phenotype of *MdABI5*-RNAi transgenic lines. Compared with the *MdABI5*-OE transgenic lines, *MdABI5*-RNAi plants were more sensitive to drought treatment (Figure 6c). Under drought treatment, *MdABI5*-RNAi lines showed lower survival rates (Figure 6d) and higher ion leakage compared with wild-type plants (Figure 6e). Analysis of the leaf relative water content and leaf dehydration in detached leaves revealed that *MdABI5*-RNAi lines had a lower leaf relative water content (Figure 6f) and higher leaf dehydration rates compared with wild-type plants (Figure 6g). Analysis of detached leaves revealed higher transpiration in *MdABI5*-RNAi leaves compared with wild-type plants. Under drought conditions, *MdABI5*-RNAi lines had lower photosynthetic rates (Figure 6h), a higher catalase (CAT) content (Figure 6i), and higher hydrogen peroxide (H_2O_2) levels compared with wild-type plants (Figure 6j), indicating that *MdABI5*-RNAi lines were more affected by drought stress compared with wild-type plants. Based on the results of our experiments with *MdABI5*-OE and *MdABI5*-RNAi, we propose that *MdABI5* plays a positive role in enhancing drought tolerance in apple.

Histone modification-regulated MdOCP3 positively regulates drought tolerance in apple

The ATAC-seq results of our study revealed significant hypo-regulation of the chromatin accessibility of the *MdOCP3* promoter and significant down-regulation of the expression of

Figure 5 Over-expression of *MdABI5* enhanced the drought tolerance in apple. (a) Expression of *MdABI5* under drought treatment with different soil water contents in *Malus hupehensis*. (b) Genome browser snapshots of H3K14ac and H3K27me3 upstream of the *MdABI5* in drought treatment and control. (c) ChIP-qPCR verification of histone modifications upstream of ABI5. (d) Identification of *MdABI5* OE transgenic lines at DNA level. M, marker; PC, positive control; WT, wild-type apple plants (GL-3); #1, #2, and #12: three *MdABI5*-OE transgenic lines 1, 2, 12. (e) Identification of *MdABI5*-OE lines at RNA level. Error bars indicate SD ($n = 3$). (f) Phenotypes of GL-3 (Wild-type) and *MdABI5*-OE transgenic plants. Bars = 5 cm. Three-month-old plants were treated under drought for 14 d and then rewatered for 7 d. (g) Survival rates of GL-3 and three independent *MdABI5*-OE lines in (f). Data are mean $\pm$ SD ($n = 3$). At least 18 seedlings were used per line. (h) Leaf electrolyte leakage assays of GL-3 and *MdABI5*-OE plants under drought treatment. Data are mean $\pm$ SD ($n = 7$). (i) Leaf relative water content (RWC) of GL-3 and *MdABI5*-OE plants under drought. Data are mean $\pm$ SD ($n = 6$). (j) Water loss of detached leaves from GL-3 and *MdABI5*-OE plants at 25 °C. Data are mean $\pm$ SD ($n = 6$). (k) Photosynthesis rate analysis of GL-3 and *MdABI5*-OE plants under drought treatment. Data are mean $\pm$ SD ($n = 9$). (l) Catalase (CAT) activity in GL-3 and *MdABI5*-OE plants under drought treatment. Data are mean $\pm$ SD ($n = 3$). (m) Hydrogen peroxide (H_2O_2) contents in GL-3 and *MdABI5*-OE plants under drought treatment. Data are mean $\pm$ SD ($n = 3$). One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.



MdOCP3. In Arabidopsis, the *ocp3* mutation positively regulates the drought response through an ABA-dependent pathway (Ramírez et al., 2009). However, no research has examined the role of *MdOCP3* in apple under drought stress. Similar to

Arabidopsis, drought stress reduces the expression of *MdOCP3*, and its expression gradually decreases with the severity of drought (Figure 7a). Upstream of *MdOCP3*, we observed a decrease in the enrichment of H3K9ac and H3K36me3, indicating

that the down-regulation of *MdOCP3* may stem from the regulation of H3K9ac and H3K36me3 (Figure 7b). Consistent with the ChIP-seq results, ChIP-qPCR verified changes in histone modifications upstream of *OCP3* (Figure 7c). To explore the role

of *MdOCP3* in apple drought resistance, we first generated two *MdOCP3* overexpression (*MdOCP3*-OE) transgenic apple lines (Figure 7d). The expression of *MdOCP3* was 40 to 50-fold higher in *MdOCP3*-OE transgenic plants than in wild type (GL-3)

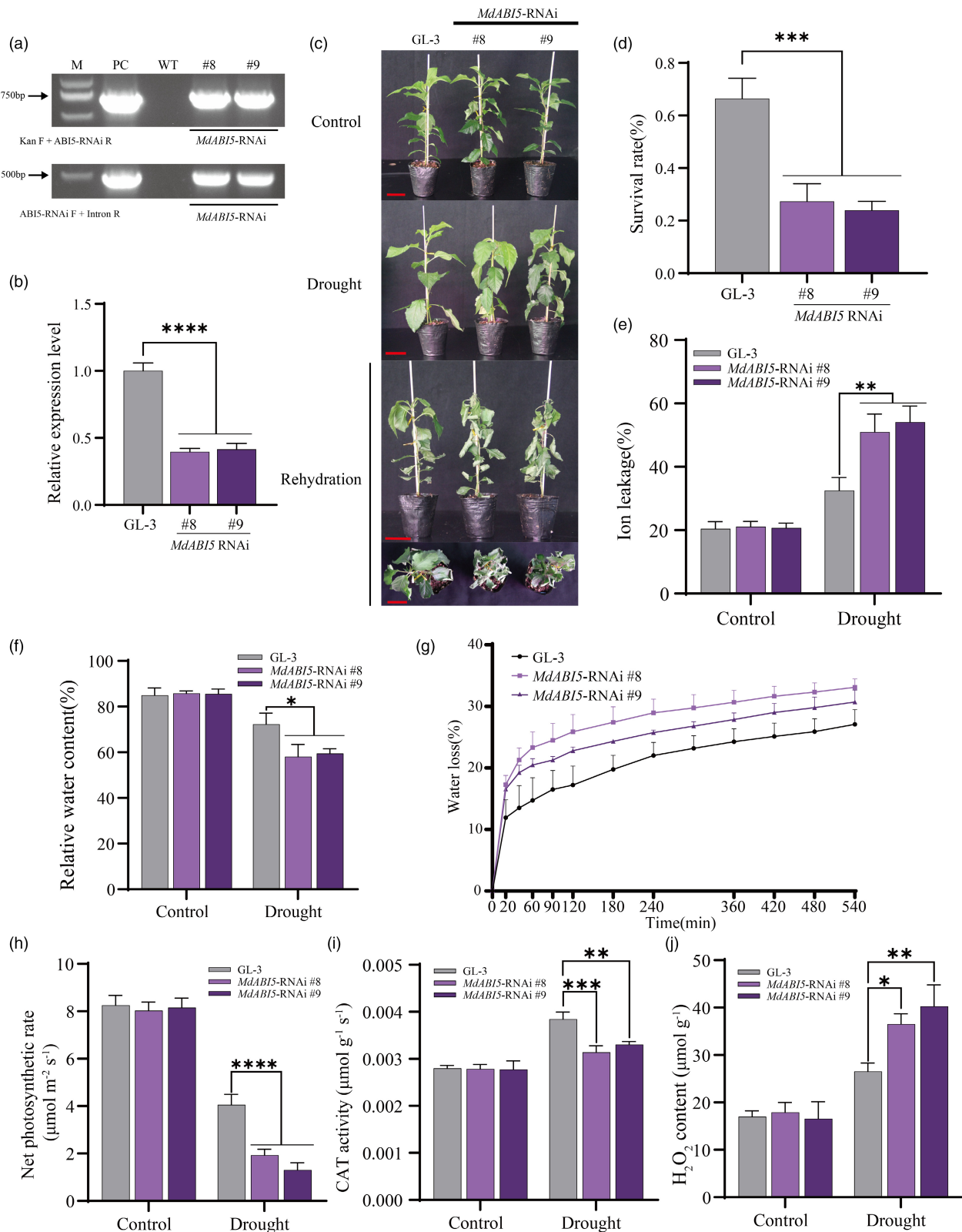


Figure 6 RNA interference of *MdABI5* decreased the drought tolerance in apple. (a) Identification of *MdABI5* RNAi transgenic lines at DNA level. M, marker; PC, positive control; WT, wild-type apple plants (GL-3); #8, #9: two *MdABI5*-RNAi transgenic lines 8, 9. (b) Identification of *MdABI5*-OE lines at RNA level. Error bars indicate SD ($n = 3$). (c) Phenotypes of GL-3 (Wild-type) and *MdABI5*-RNAi transgenic plants. Bars = 5 cm. Three-month-old plants were treated under drought for 14 d and then rewatered for 7 d. (d) Survival rates of GL-3 and two independent *MdABI5*-RNAi lines in (c). Data are mean $\pm$ SD ($n = 3$). At least 15 seedlings were used per line. (e) Leaf electrolyte leakage assays of GL-3 and *MdABI5*-RNAi plants under drought treatment. Data are mean $\pm$ SD ($n = 4$). (f) Leaf relative water content (RWC) of GL-3 and *MdABI5*-RNAi plants under drought. Data are mean $\pm$ SD ($n = 4$). (g) Water loss of detached leaves from GL-3 and *MdABI5*-RNAi plants at 25 °C. Data are mean $\pm$ SD ($n = 3$). (h) Photosynthesis rate analysis of GL-3 and *MdABI5*-RNAi plants under drought treatment. Data are mean $\pm$ SD ($n = 9$). (i) Catalase (CAT) activity in GL-3 and *MdABI5*-RNAi plants under drought treatment. Data are mean $\pm$ SD ($n = 3$). (j) Hydrogen peroxide (H_2O_2) contents in GL-3 and *MdABI5*-RNAi plants under drought treatment. Data are mean $\pm$ SD ($n = 3$). One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

(Figure 7e). Under short-term drought stress, *MdOCP3*-OE lines were more drought tolerant than wild type lines, which is in contrast to results in *Arabidopsis* (Figure 7f). The high survival rate of *MdOCP3*-OE lines after rehydration for 7 days indicates that its drought stress tolerance was enhanced (Figure 7g). Data on ion leakage and the relative leaf water content (RWC) after drought stress revealed that *MdOCP3*-OE lines had a lower ion leakage (Figure 7h) and higher RWC (Figure 7i) than wild type plants, suggesting that *MdOCP3*-OE lines are less affected by drought stress. Similarly, analysis of detached leaf water loss also confirmed that *MdOCP3*-OE leaves had a lower rate of water loss compared with wild type plants (Figure 7j). Additional studies of the internal environment of apple plants revealed that *MdOCP3*-OE lines had a higher net photosynthetic rate (Figure 7k) and increased catalase (CAT) (Figure 7l) and peroxidase (POD) (Figure 7m) activity after drought stress, which indicates that *MdOCP3*-OE has better drought tolerance and is less affected by drought stress compared with the wild type plants. The malondialdehyde (MDA) content (Figure 7n) and diaminobenzidine (DAB) staining (Figure 7o) showed that the drought treatment induced less cell membrane damage and hydrogen peroxide (H_2O_2) accumulation in *MdOCP3*-OE plants compared with wild type plants. These results indicate that *MdOCP3*-OE plants have strong drought resistance based on their lower water loss rate and the stability of the internal environment under drought conditions.

To further confirm the function of *MdOCP3* after drought treatment, we constructed *MdOCP3* RNA-interfering (*MdOCP3*-RNAi) transgenic apple plants and performed a phenotypic analysis under drought treatment (Figure 8a,b). In contrast to the *MdOCP3*-OE lines, *MdOCP3*-RNAi lines were sensitive to drought compared with wild-type plants (Figure 8c). *MdOCP3*-RNAi lines had a lower survival rate (Figure 8d) and higher ion leakage (Figure 8e). The leaf relative water content and water loss from detached leaves indicated that *MdOCP3*-RNAi lines had lower water retention (Figure 8f) and correspondingly faster rates of leaf water loss (Figure 8g). The leaf photosynthetic rate was significantly lower in *MdOCP3*-RNAi plants under drought stress than in wild-type plants (Figure 8h). Detection of enzymes and substances related to the environment in apple lines suggested that *MdOCP3*-RNAi lines had lower catalase (CAT) (Figure 8i) and peroxidase (POD) (Figure 8j) activities and a higher MDA content (Figure 8k). Diaminobenzidine (DAB) staining (Figure 8l) showed that *MdOCP3*-RNAi accumulated more hydrogen peroxide (H_2O_2) after drought treatment compared with wild-type plants. Studies of *MdOCP3*-OE revealed that *MdOCP3* positively regulates drought tolerance in apple.

Effects of epigenetic modifications in drought regulatory networks

Research on DEGs has revealed that many drought-responsive genes are enriched in different epigenetic modifications. However, the role of epigenetic modifications in the regulatory network underlying the drought response requires further study. To clarify the regulatory role of the epigenetic modifications in drought response pathways in apple, we annotated changes in epigenetic modifications through study of genes known to be involved in the drought response network (Liu et al., 2023; Zhang et al., 2022). Members of the ABA signalling response pathway were regulated by different epigenetic modifications, including key genes encoding PYLs, PP2C, SnRKs, and RAFs (Figure 9a). These genes play important roles by responding to ABA and protein phosphorylation, which affects the downstream regulatory roles of drought-related transcription factors (Liu et al., 2023; Zhang et al., 2022). This demonstrates that epigenetic modifications play a crucial role in the ABA response and signal transduction. Genes related to calcium ion pathways and kinase cascades are also regulated by epigenetic modifications, including CIPKs and CPKs, which play important roles in drought signalling transduction and stomatal closure (Liu et al., 2023; Zhang et al., 2022). In addition, genes such as *OST1*, *NCED3*, and *WRKY53* in the ABA-dependent pathway (Figure 9a,b) and genes such as *MYB88* and *CBFs* in the ABA-independent pathway (Figure 9c) are also regulated by epigenetic modifications. These drought-related transcription factors can also affect the phenotype by regulating downstream drought-responsive genes to further regulate the drought response in apple.

We also studied genes regulated by multiple histone modifications to reveal the role of complex histone-regulated genes under drought treatment. During the drought response, genes involved in the response to drought stress, such as *bHLH14-like*, *MYC2-like*, *MYC3-like*, *DREB1D*, *HAT5*, *ERF113*, and *ERF061*, are influenced by single histone modifications. Genes such as *WRKY11*, *DREB1A-like*, and *DREB1B* are regulated by more than two types of histone modifications. DNA methylation changes upstream of *bHLH35-like*, *SCL14*, and *YABBY2* can also affect the expression of these genes (Figure 10a, Figure S3).

To verify the reliability of the results, we performed the drought treatment again and recorded differences in the soil water content, such as a soil water content of 60% after drought treatment (referred to as S60d, S30d, S10d, and S5d). The RT-qPCR results of the transcription factors, epigenetic factors, and DNA methylation factors were consistent with transcriptomic

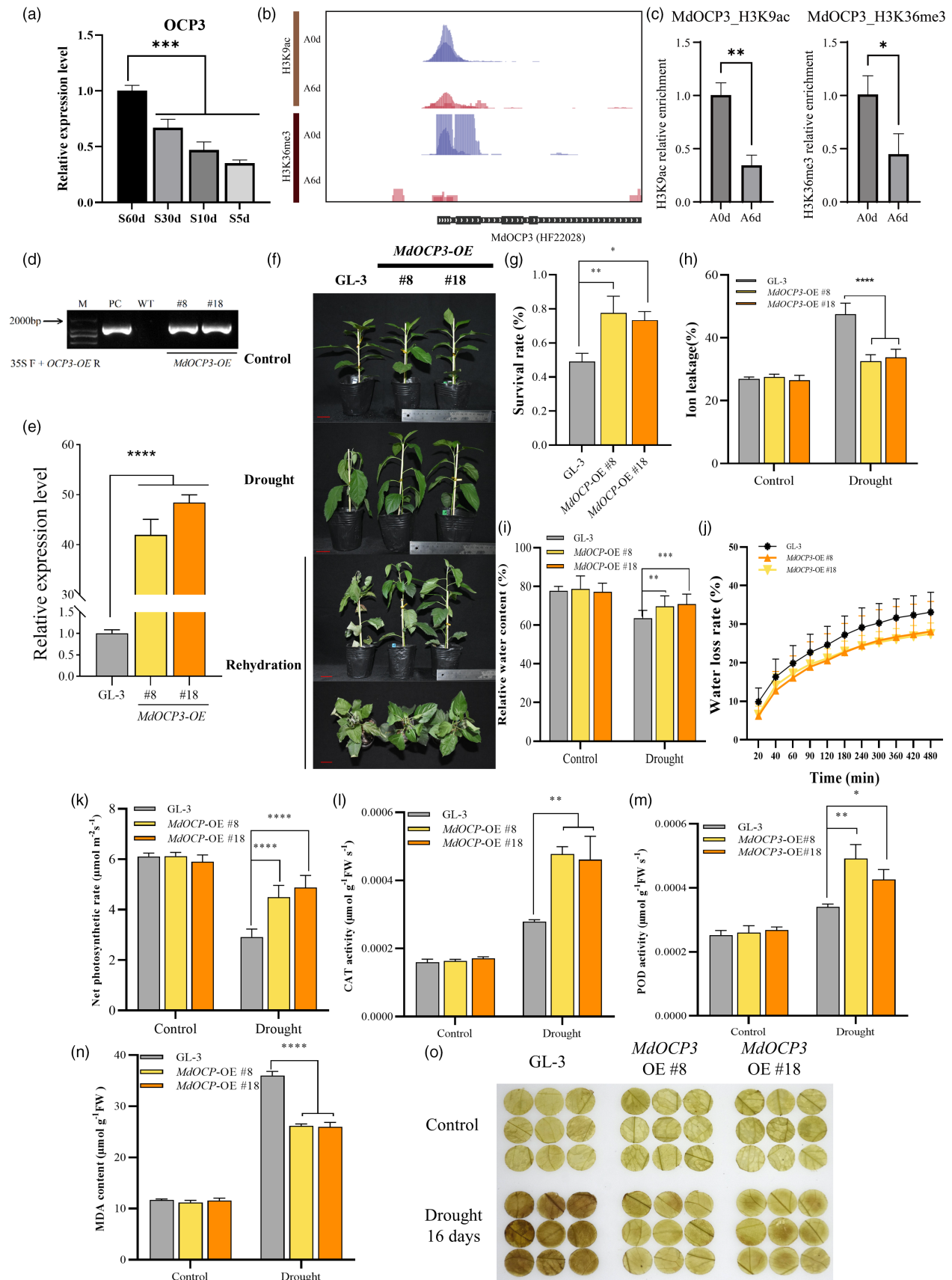


Figure 7 Over-expression of *MdOCP3* enhanced the drought tolerance in apple. (a) Expression of *MdOCP3* under drought treatment with different soil water contents in *Malus hupehensis*. Error bars indicate SD ($n = 3$). (b) Genome browser snapshots of H3K9ac and H3K36me3 upstream of the *MdOCP3* in drought treatment and control. (c) ChIP-qPCR verification of histone modifications upstream of *OCP3*. (d) Identification of *MdOCP3*-OE transgenic lines at DNA level. M, marker; PC, positive control; WT, wild-type apple plants (GL-3); #8, #18: two *MdOCP3*-OE transgenic lines 8, 18. (e) Identification of *MdOCP3*-OE lines at RNA level. Error bars indicate SD ($n = 3$). (f) Phenotypes of GL-3 (Wild-type) and *MdOCP3*-OE transgenic plants. Bars = 3 cm. Three-month-old plants were treated under drought for 18 d and then rewatered for 7 day. (g) Survival rates of GL-3 and two independent *MdOCP3*-OE lines in (f). Data are mean $\pm$ SD ($n = 3$). At least 18 seedlings were used per line. (h) Leaf electrolyte leakage assays of GL-3 and *MdOCP3*-OE plants under drought treatment. Data are mean $\pm$ SD ($n = 8$). (i) Leaf relative water content (RWC) of GL-3 and *MdOCP3*-OE plants under drought. Data are mean $\pm$ SD ($n = 8$). (j) Water loss of detached leaves from GL-3 and *MdOCP3*-OE plants at 25 °C. Data are mean $\pm$ SD ($n = 10$). (k) Photosynthesis rate analysis of GL-3 and *MdOCP3*-OE plants under drought. Data are mean $\pm$ SD ($n = 9$). Detection of CAT (l), POD (m), and MDA (n) contents in GL-3 and *MdOCP3*-OE plants under drought treatment. Data are mean $\pm$ SD ($n = 3$). (o) DAB staining was performed to detect the H_2O_2 contents of GL-3 and *MdOCP3*-OE. CAT, catalase; POD, peroxidase; MDA, malondialdehyde; H_2O_2 , hydrogen peroxide; DAB, 3,30-DAB tetrahydrochloride. One-way ANOVA (Tukey's test) was performed and statistically significant differences were indicated by * $P \leq 0.05$; ** $P \leq 0.01$; *** $P \leq 0.001$; **** $P \leq 0.0001$.

results. MYC3-like, HAT5, and bHLH14-like may be affected by H3K4me3 and H3K27me3, and their expression was significantly up-regulated after drought treatment. The expression of MYC2-like, WRKY11B, DREB1A-like, and DREB1B may be down-regulated by histone modifications. The epigenetic modifier factors AGO2, ATX2, and ROS1-like may also regulate the enrichment of epigenetic modifications at specific genomic loci in response to drought (Figure 10b).

Discussion

Epigenetic modifications play an important role in regulating the response to abiotic stress (Lloyd and Lister, 2022; Ueda and Seki, 2020). Clarifying changes in epigenetic modifications during drought stress is important in studies of the molecular mechanisms of drought resistance in apples. We analysed the transcriptome, DNA methylation, and six histone modifications in apples after drought treatment for 0, 3, 6, and 9 days. The results revealed changes in DNA methylation and histone modifications at the whole-genome level after drought treatment. We verified the functions of two drought-responsive candidate genes *MdABI5* and *MdOCP3* regulated by histone modifications in the response of apple to drought.

Based on the transcriptome analysis, 3041 DEGs were identified in apple leaves after 3 days of drought stress, and GO enrichment analysis revealed that these DEGs were strongly correlated with water deprivation. The number of DEGs was highest in the A6d versus A0d comparison (Figure 1b). Many genes involved in DNA methylation and histone modification pathways were most highly expressed in A6d, and these genes were down-regulated in A9d (Figures 2e and 3b). The most marked changes in expression occurred before observed phenotypic changes in leaves under drought treatment, which indicates that A6d is a critically important time in the drought response (Cao et al., 2022). DNA methylation plays a crucial role in maintaining genome stability and gene silencing (Zhang et al., 2018). The levels of three types of DNA methylation increased under drought treatment in apple. Drought treatment does not always induce an increase in DNA methylation levels in different plants. The magnitude of the down-regulation of mCG, mCHG, and mCHH in the promoter region varies in strawberry (Cao et al., 2022). Three types of DNA methylation in the promoter region are up-regulated in Arabidopsis under drought (Ganguly et al., 2017). The average methylation levels of the CG and CHG contexts in sea buckthorn leaves increase under drought stress. However, the methylation level of the CHH context was not observed to change (Lyu et al., 2023). In tea

plants, DNA methylation is significantly up-regulated under drought conditions (Yue et al., 2023). Under drought treatment, the methylation levels of mCG and mCHG in the apples 'Qinguan' and 'Honeycrisp' slightly increase, and the methylation level of mCHH decreases (Xu et al., 2018). The changes in DNA methylation under drought treatment may differ among plants and varieties. This could reflect variation in the regulatory mechanisms of DNA methylation among plants under drought. The abundance of DMRs peaked at A3d, indicating that DNA methylation occurs prior to transcriptional changes in the response to drought. The discrepancy between DNA methylation and transcription levels may be explained by the limited overlap between DMRs and DEGs near gene regions. The significant differences in DMRs suggest that DNA methylation may play a crucial role in the response to drought. In addition, differences in levels of mCpG, mCHG, and mCHH modifications have been observed in various genes (Figure 2a). Previous studies have reported positive correlations between DNA methylation of the gene body and gene expression (Liang et al., 2014; Xu et al., 2018). Further research is needed to investigate the role of DNA methylation of the gene body in the drought response. Aside from direct changes in methylation levels and DMRs, previous studies have indicated that RdDM components play a key role in the drought response, and our data revealed the differential expression of RdDM-related genes, such as DCL3 (Figure 2e) (Huang et al., 2016b). However, we did not observe significant differences in the enrichment of 24-nt sRNAs near gene regions after drought treatment (Figure 2d). Further differential analysis of 24-nt siRNAs for specific genes could shed light on the role of 24-nt siRNAs in the drought stress response. Based on changes in the expression of genes related to 24-nt sRNAs, changes in mCHH in the gene region after drought may be caused by demethylation pathways (Figure 2a,d,e). After drought treatment, hyper-regulation of the gene body DNA methylation level was observed (Figure 2a), and gene body histone modifications appeared to slightly decrease after drought treatment (Figure 3a). However, in our previous study, ATAC-seq data for GL3 after drought treatment revealed a significant increase in chromatin accessibility near the gene regions (Wang et al., 2022a), indicating that the epigenetic process of chromatin opening during drought is complex.

Previous studies have found that genes may be synergistically regulated via various histone modifications and DNA methylation. The synergy among epigenetic modifications plays an important role in the transcriptional activation, repression, and bivalent regulation of genes (Ruthenburg et al., 2007). At the whole-genome level, the distribution of H3K9ac and H3K14ac is similar

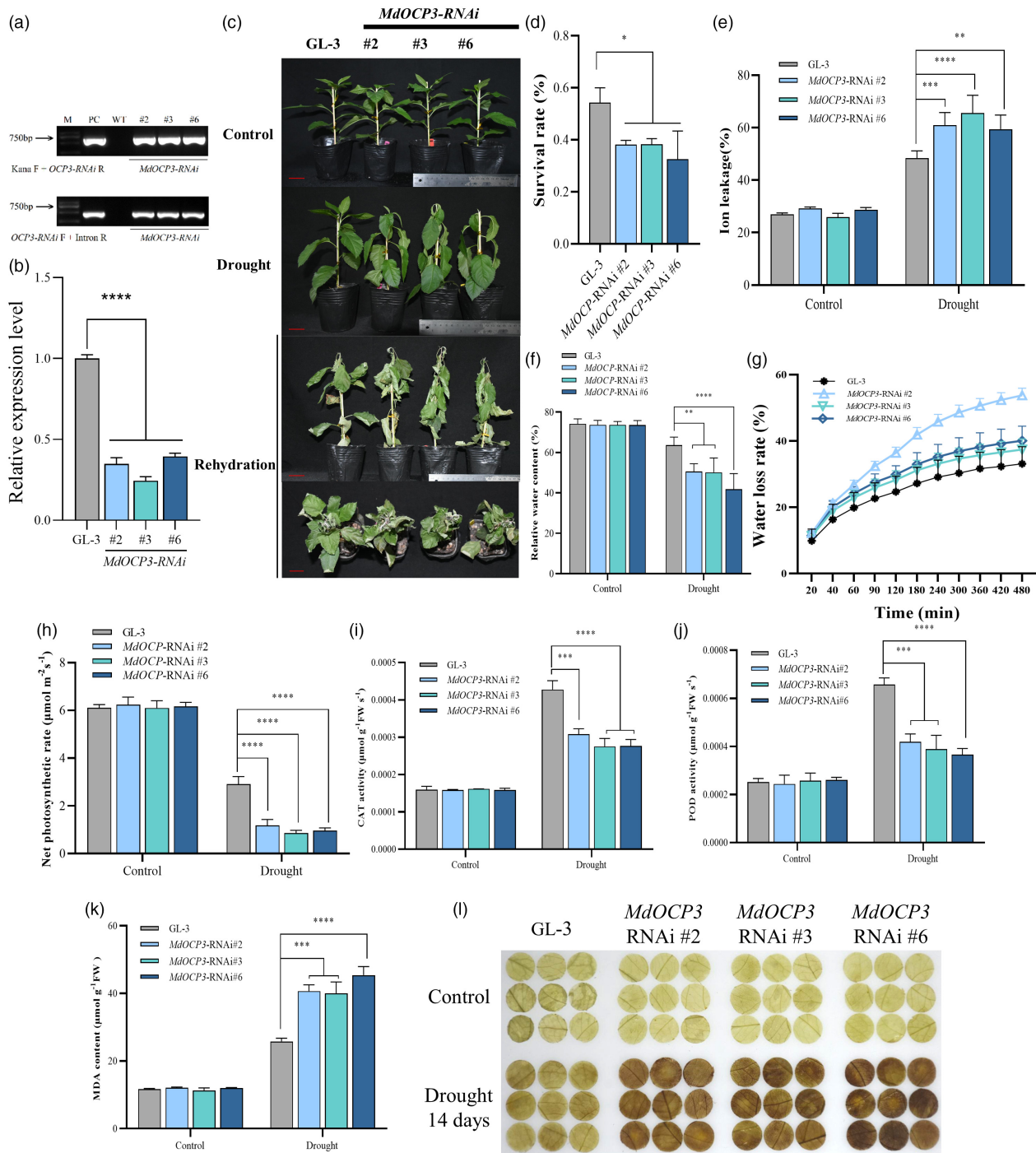


Figure 8 RNA interference of *MdOCP3* decreased the drought tolerance in apple. (a) Identification of *MdOCP3*-RNAi transgenic lines at DNA level. M: marker; PC: positive control; WT: wild-type apple plants (GL-3); #2, #3, #6: three *MdOCP3*-RNAi transgenic lines 2, 3, 6. (b) Identification of *MdOCP3*-RNAi lines at the RNA level. Error bars indicate SD ($n = 3$). (c) Phenotypes of GL-3 (Wild-type) and *MdOCP3*-RNAi transgenic plants. Bars = 3 cm. Three-month-old plants were treated under drought for 15 d and then rewatered for 7 d. (d) Survival rates of GL-3 and three independent *MdOCP3*-RNAi lines in (c). Data are mean $\pm$ SD ($n = 3$). At least 18 seedlings were used per line. (e) Leaf electrolyte leakage assays of GL-3 and *MdOCP3*-RNAi plants under drought treatment. Data are mean $\pm$ SD ($n = 8$). (f) Leaf relative water content (RWC) of GL-3 and *MdOCP3*-RNAi plants under drought. Data are mean $\pm$ SD ($n = 8$). (g) Water loss of detached leaves from GL-3 and *MdOCP3*-RNAi plants at 25 °C. Data are mean $\pm$ SD ($n = 10$). (h) Photosynthesis rate analysis of GL-3 and *MdOCP3*-RNAi plants under drought. Data are mean $\pm$ SD ($n = 9$). Detection of CAT (i), POD (j), and MDA (k) contents in GL-3 and *MdOCP3*-RNAi plants under drought treatment. Data are mean $\pm$ SD ($n = 3$). (l) DAB staining was performed to detect the H_2O_2 contents of GL-3 and *MdOCP3*-RNAi. CAT, catalase; POD, peroxidase; MDA, malondialdehyde; H_2O_2 , hydrogen peroxide. DAB, 3,30-DAB tetrahydrochloride. One-way ANOVA (Tukey's test) was performed, and statistically significant differences were indicated by * $P \leq 0.05$; ** $P \leq 0.01$; *** $P \leq 0.001$; **** $P \leq 0.0001$.

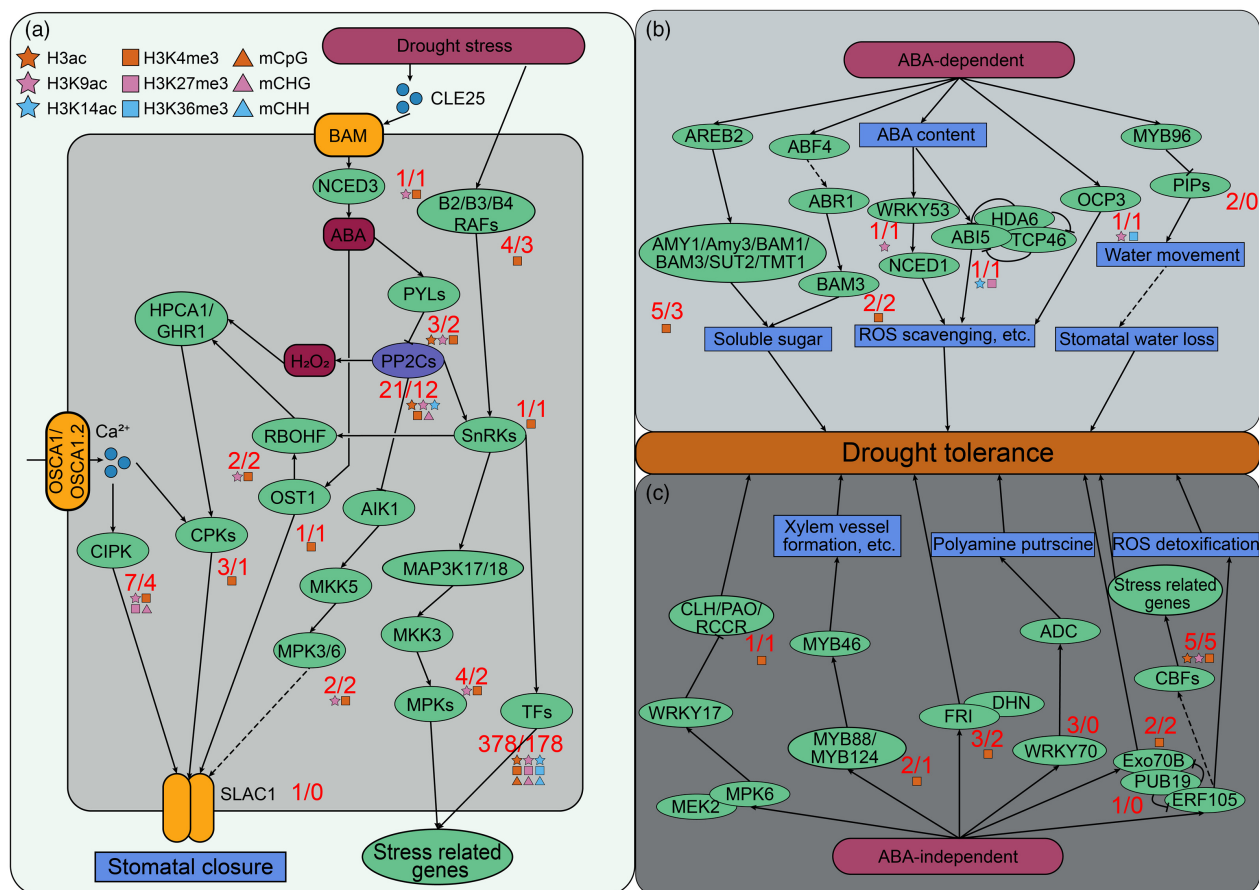


Figure 9 Sensing, signalling, and ABA-dependent and non-dependent transcriptional regulatory networks for drought stress in plants. Drought response pathways are referenced in reported reviews (Liu *et al.*, 2023). (a) Epigenetic modification annotation of factors related to drought stress sensing and signalling in plants. The number of TFs is all the TFs differentially expressed throughout the drought, not just the TFs regulated by SnRKs. (b, c) ABA-dependent and ABA-independent regulatory networks annotated by epigenetic modifications. Star shapes represent histone acetylation modified DEGs, squares represent histone methylation modified DEGs, and triangles represent DNA methylation modified DEGs. Different colours represent different modification types.

to overall acetylation levels (H3ac), suggesting that they are potentially correlated with histone acetylation in apple. Although there were differences in the distributions of histone acetylation and histone methylation, the distribution of H3K4me3 was similar to that of histone acetylation at many locations, indicating that H3K4me3 may play a role in the activation of genes that may be co-regulated by histone acetylation. Although the distributions of H3K27me3 and H3K4me3 are similar in certain regions, the overall distribution of these histone modifications differs, suggesting that the bivalent effect of H3K27me3 and H3K4me3 in the apple genome is limited. H3K36me3 has a relatively unique distribution in certain regions, which suggests that it plays a unique regulatory role in these regions (Figure S2C). There was no significant overlap in DHMRs between activating and repressive histone modifications observed at A0d and A6d (Figure 4a). In drought-induced up-regulated genes, the activating marks H3K4me3 appear to be associated with genes with lower fold changes, and the hypo-regulation of H3K27me3 is related to up-regulated genes with higher fold changes (Figure 4a). This preference is not present in down-regulated genes, and no preference in the fold change of DMR-related genes was observed. Changes in H3K27me3 in down-regulated genes are mostly positively correlated with gene regulation, which may

indicate that H3K27me3 does not play a major role in the regulation of down-regulated genes during drought stress (Figure 4a). Our results showed that the distribution of DNA methylation is generally opposite to the distribution of genes and histone modifications in apple. Although previous studies suggest an indirect relationship between H3K27me3 and DNA methylation (Lindroth *et al.*, 2004), the correlation between H3K27me3 and DNA methylation at the genomic level appears relatively weak in apple (Figure S2C,D). However, the distribution of DHMRs with the H3K27me3 modification and DMRs with DNA methylation showed partial overlap compared with other types of modifications (Figure 4a), suggesting that there is a potential correlation between H3K27me3 and DNA methylation in apple in response to drought stress. The significant enrichment of pathways related to growth and development was observed in the enrichment results of DEGs, DMRs, and DHMRs (Figures 1e, 3d). This may indicate that DNA methylation and histone modifications play an important role in balancing energy allocation between drought response and development processes.

ABI5 plays an important role in regulating the ABA response pathway by targeting PYLs, which contributes to plant drought tolerance (Zhao *et al.*, 2020). *MdABI5* has been reported to play a

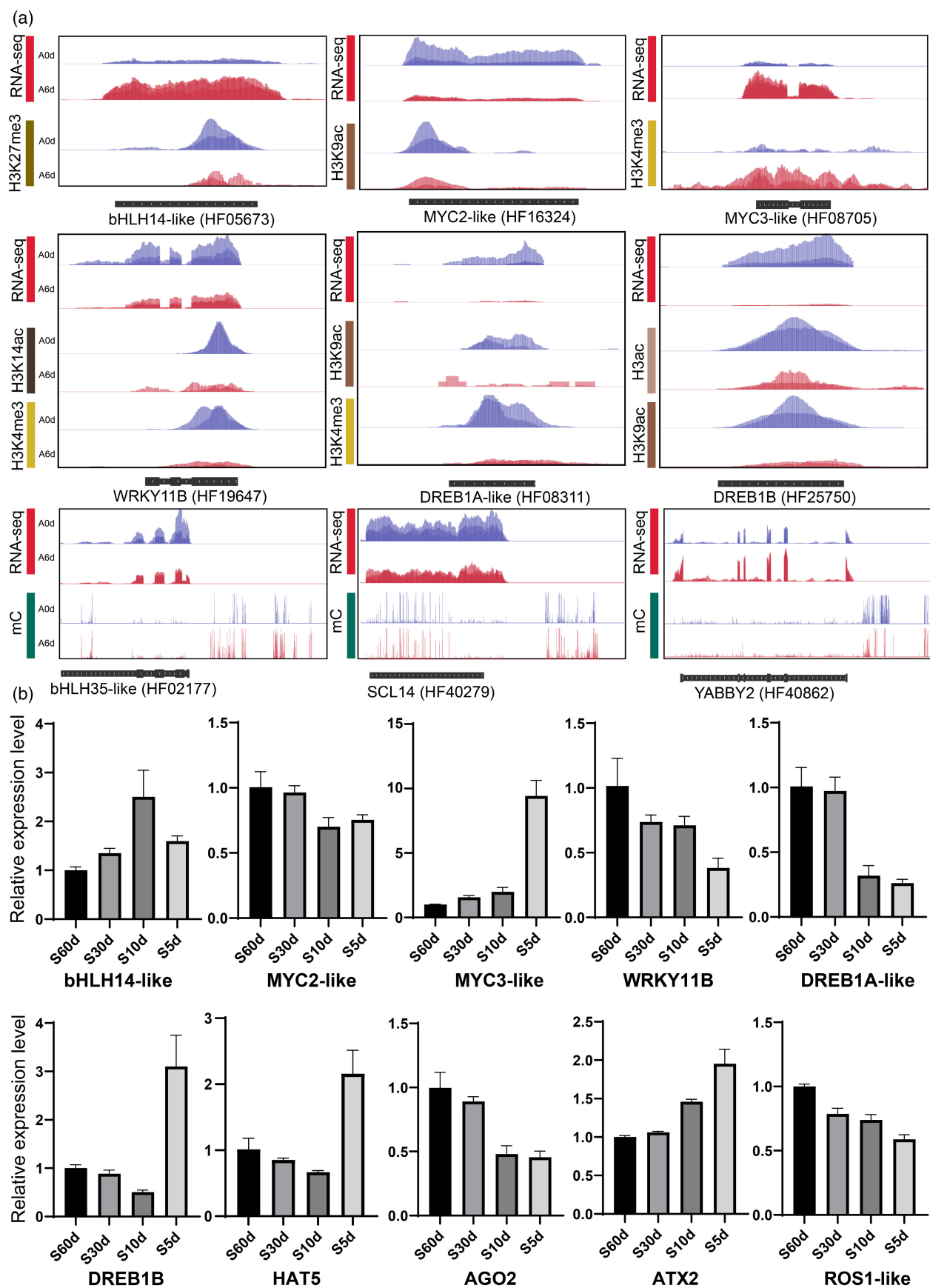


Figure 10 Genomic snapshots and expression validation of epigenetic factors and epigenetically regulated genes. (a) Genome browser snapshots showing the relationship between transcript levels with DNA methylation and histone modifications in the 6-day drought treatment and control. (b) The RT-qPCR result of ten differential expressions of genes regulated by epigenetic modifications and drought-responsive epigenetic factors under the different soil water content drought treatment in *Malus hupehensis*. Error bars indicate SD ($n = 3$).

role in the drought response process and is negatively regulated by *MdTCP46* (Liu et al., 2022). *MdABI5* interacts with *MdHDA6* to regulate the histone acetylation of specific sites, which in turn regulates the expression of genes (Li et al., 2023b). In our study, we found that the histone activation mark H3K14ac was hyper-regulated and the repressive mark H3K27me3 was hypo-regulated upstream of *MdABI5*. However, we did not observe any changes in chromatin accessibility upstream of *MdABI5* in our previous study (Wang et al., 2022a). To characterize whole-plant *MdABI5* drought resistance, we utilized leaf disc transformation to obtain transgenic plants and evaluate the drought resistance of *MdABI5*. We found that *MdABI5* positively regulates the drought resistance of apples.

OCP3 is a homeodomain (HD) family transcription factor. In Arabidopsis, *ocp3* mutants regulate the resistance of plants through ABA and JA (García-Andrade et al., 2011; Ramírez et al., 2009). However, no studies have clarified the role of *MdOCP3* in the drought resistance of apples. Our findings indicated that the upstream histone activation marks H3K14ac and H3K36me3 of *MdOCP3* were hypo-regulated, and a decrease in chromatin accessibility in this region was observed (Wang et al., 2022a), indicating that histone modifications may regulate the expression of *MdOCP3* under drought stress by influencing chromatin accessibility in this region. However, drought resistance is enhanced and reduced in *MdOCP3*-OE and *MdOCP3*-RNAi plants, respectively. The reasons underlying the down-regulation of *MdOCP3* under drought stress and the functional differences in the *OCP3* gene between Arabidopsis and apple require further investigation.

Our findings revealed changes in the whole-genome non-coding RNA, histone modifications, and DNA methylation after different drought treatments. We found that the response to drought stress was most significant after 6 days of drought treatment in apple. The level of DNA methylation continuously changed with the severity of drought stress, peaking at A3d. Histone modifications, including H3K9ac, H3K14ac, H3K4me3, H3K27me3, and H3K36me3, are involved in the response to drought and the regulation of gene expression. Up-regulated genes with higher fold change under drought stress may be mainly attributed to the hypo-regulation of H3K27me3, and up-regulated genes with a lower fold change may be regulated by H3K4me3. Phenotypic experiments demonstrated that both *MdABI5* and *MdOCP3*, regulated by histone modifications, positively contribute to drought tolerance in apple. Our findings provide new insights with implications for studies of the epigenetic landscape in apple under drought stress and will aid future epigenetic studies under drought stress.

Material and methods

Plant materials and drought treatment

The material used for drought treatment was the 3-month seedling leaves of *Malus hupehensis*. The experiments were conducted at Northwest A&F University, Yangling, located in the loess plateau of China (34°200' N, 108°240' E). All plants were

grown in the greenhouse (temperature 25–30 °C, humidity 50%). Plants of similar height and growth conditions were selected and cultivated in the same weight of substrate (the ratio of peat, perlite, and vermiculite is 3/1/1, v/v/v). Before drought treatment, the soil was kept at saturated moisture and left for 2 days, which was recorded as Day 0 of the drought. The top 4th–6th leaves of each of the three independent *Malus hupehensis* plants were mixed as a biological replicate. Leaves were taken after 0 days of drought (A0d), 3 days of drought (A3d), 6 days of drought (A6d), and 9 days of drought (A9d), kept immediately in liquid nitrogen or formaldehyde, and stored at –80 °C. Samples of *Malus hupehensis* leaves from various drought days were used for subsequent RNA-seq, WGBS, and ChIP-seq analysis.

In the same experimental condition, to ensure an accurate assessment of the level of drought stress and validate the robustness of our results, we collected the 4th–6th leaves of *Malus hupehensis* with a soil moisture content of 60% after drought treatment (S60d), 30% (S30d), 10% (S10d), and 5% (S5d). The selection of these moisture levels was based on the saturated moisture content as a reference criterion. Subsequently, the samples were rapidly frozen in liquid nitrogen and stored at –80 °C. The soil of the sampled plants was quickly mixed after sampling and weighed to take 5 g in an aluminium box, which was placed in an oven at 105 °C for 12 h, cooled in a desiccator, and weighed to calculate the water content. The percentage of soil moisture content at different drought treatments is defined as the ratio of soil water content to the saturated water content at the time point. Samples of *Malus hupehensis* leaves from various drought treatments were used for subsequent RNA extraction and RT-qPCR.

The material used for the genetic transformation of apples was the tissue culture leaves of apple line GL-3, derived from the 'Royal Gala' (*Malus × domestica*) variety. This line GL-3 was selected due to its high regeneration and transgenic efficiency for apple genetic transformation (Dai et al., 2013). *MdABI5*, *MdOCP3*, and GL-3 lines underwent a selection process, subculturing, rooting, and transplantation during cultivation (Geng et al., 2018; Li et al., 2023b). Subsequently, they underwent a short-term drought treatment, involving 14–18 days of water deprivation in the plant culture rooms (25 °C, 60% relative humidity, and 16 h-light/8 h-dark).

Library construction and analysis process for ssRNA-seq samples

Twelve samples were prepared for subsequent whole-transcriptome analysis. RNA degradation and contamination were assessed using 1% agarose gel. The purity, concentration, and integrity of the total RNA were evaluated using the NanoPhotometer® spectrophotometer (IMPLEN, CA, USA), the Qubit® RNA Assay Kit with the Qubit® 2.0 Fluorometer (Life Technologies, CA, USA), and the RNA Nano 6000 Assay Kit on the Bioanalyzer 2100 system (Agilent Technologies, CA, USA).

RNA sample preparations utilized a total of 3 µg RNA per sample as input material. Initially, the Epicentre Ribo-zero™ rRNA

Removal Kit (Epicentre, USA) was employed to eliminate ribosomal RNA. Subsequently, rRNA-depleted RNA was utilized to generate 12 sequencing libraries using the NEBNext® Ultra™ Directional RNA Library Prep Kit for Illumina® (NEB, USA). For small RNA sequencing, the same RNA per sample was used as input material, and libraries were constructed using the NEBNext® Multiplex Small RNA Library Prep Set for Illumina® (NEB, USA).

Following the instructions of the manufacturer, the clustered index-coded samples were processed on the cBot Cluster Generation System using the TruSeq PE Clustering Suite v3-cBot-HS (Illumina). Subsequently, the Illumina HiSeq 4000 platform was utilized for sequencing the libraries and generating 150 bp paired-end reads.

The ssRNA-seq process includes quality control, sequence alignment, and differential analysis. Clean data were obtained by removing adapters, poly-N-containing reads, and low-quality reads from the raw data. Clean data were used for quality control by FastQC v0.11.9 (<http://www.bioinformatics.babraham.ac.uk/projects/fastqc/>) with default parameters. After quality control, the clean reads were mapped to the reference genome (Zhang *et al.*, 2019) by HISAT2 v2.1.0 (Kim *et al.*, 2019). After the alignment, use samtools (v1.9) with parameter -bq 1 (Li *et al.*, 2009) to get the sorted bam file. StringTie (v2.2.1) was used to assemble the lncRNA transcriptions and counted the known mRNAs (Pertea *et al.*, 2015). Transcripts from each strand were merged using GffRead (v0.12.7) and GffCompare (v0.12.6) with the default setting (Pertea and Pertea, 2020). Different classes of lncRNAs (such as lincRNAs, intronic lncRNAs, antisense lncRNAs, and sense lncRNAs) were identified using GFFcompare (v0.12.6). Three computational approaches were used to predict the coding potential prediction, respectively, CNCI (Coding-Non-Coding-Index) (v2), CPC (Coding Potential Calculator) (0.9-r2), and CPAT (v3.0.4) (Kong *et al.*, 2007; Sun *et al.*, 2013; Wang *et al.*, 2013). The overlap of the three software prediction results and transcript length > 200 bp for potential lncRNAs. Finally, the predicted lncRNAs were identified by comparing with databases to remove the transcription that was not lncRNAs, involving Pfam (<http://pfam.xfam.org/>), Rfam (<http://rfam.xfam.org/>), miRbase (<http://www.mirbase.org/>). To identify significant differences in lncRNA and mRNA expression, a differential expression analysis of the read counts matrix from 12 samples was conducted using R/DESeq2 (Love *et al.*, 2014). Differential gene clustering was Fuzzy C-Means Clustering using Mfuzz (Kumar and Futschik, 2007). The absolute value of $\log_2(\text{fold change}) > 0.4$ with $P < 0.05$ was defined as the different expression mRNA and lncRNA. miRNA prediction by sRNAmirer (v1.1) (Li *et al.*, 2024). 24 nt siRNA prediction and mapping reference to previous studies (Cheng *et al.*, 2018). The prediction of regulatory approaches and co-expression network construction of lncRNAs are mentioned in our previous studies (Wang *et al.*, 2022b).

Library construction and analysis process for ChIP-seq and WGBS

Chromatin immunoprecipitation (ChIP) experiment was performed as described in the previous study (Xie *et al.*, 2018). In brief, leaves from *Malus hupehensis* under different drought treatments were sliced, cross-linked in 1% (w/v) formaldehyde for 20 min, and quenched with 0.125 M glycine. Chromatin extraction was performed, followed by sonication to fragment the DNA into 0.2–0.5 kb pieces. Immunoprecipitation was performed using different histone antibodies (Millipore). The immunoprecipitated complexes were captured using ChIP-grade

protein A/G agarose beads, washed four times with appropriate buffers, and subsequently the immunoprecipitated DNA was obtained by reverse cross-linking and protein digestion. This DNA was utilized for downstream Illumina high-throughput sequencing. The WGBS experiment was conducted following the same methods as in the previous study (Xu *et al.*, 2018). ChIP-seq and WGBS library construction and analysis were performed as previously described (Li *et al.*, 2022; Song *et al.*, 2024).

Analysis of GO function enrichment and plotting

Enrichment analysis of DEGs, DARs, and DMHs results by Gene Ontology (GO) is implemented by agriGO v2.0 (<http://systemsbiology.cau.edu.cn/agriGOv2/>). Differentially expressed genes were identified as highly enriched by GO keywords with corrected P -values less than 0.05. The heatmaps were made by TBtools (Chen *et al.*, 2020). The scatter plots of GO and circos were made by using the OmicStudio tools at <https://www.omicstudio.cn/tool> (Lyu *et al.*, 2023).

RNA extraction and RT-qPCR analysis

RNAprep Pure Plant Kit (Polysaccharides & Polyphenolics-rich, DP441) from Tiangen Biotech, Beijing, China, was employed for RNA extraction from 12 drought-treated samples. Primer-BLAST (<https://www.ncbi.nlm.nih.gov/tools/primer-blast/>) was utilized to design the primers used in the experiment. The HiScript® II 1st Strand cDNA Synthesis Kit (+gDNA wiper) from Vazyme, Nanjing, China, was utilized for first-strand cDNA synthesis. Real-time quantitative PCR (RT-qPCR) analysis was conducted on a Bio-Rad CFX96TM system using ChamQ SYBR qPCR Master Mix (Vazyme, Nanjing, China).

Plasmid construction, and transgenic plant generation

The plasmid construction and transgenic plant generation were carried out as previously described (Li *et al.*, 2023a). In summary, the full-length cDNAs of *MdABI5* and *MdOCP3* were cloned into pGWB414, while CDS-specific interference fragments below 300 bp were cloned into the RNAi vector PK7GWIWG2D. The constructs were transformed into GL-3 using the leaf disc transformation method mediated by *Agrobacterium* EHA105 (Dai *et al.*, 2013). Several plant lines were obtained through resistance screening. The identified plant lines, confirmed by PCR (DNA level) and RT-qPCR (RNA level), were selected for further research.

Physiological phenotypic analysis of transgenic plants and controls under drought stress

MdABI5 and *MdOCP3* transgenic plants and controls (GL-3) were under drought treatment for 14–18 days in plant culture rooms (25 °C, 60% relative humidity, and 16 h-light/8 h-dark). During the drought treatment, phenotypic indicators were assessed, and the status of intact plants was photographed. Following the drought treatment, the wilted plants were rehydrated for 7 days, and the survival rate was subsequently calculated. For the survival rate, each treatment involved a minimum of six plants per biological replicate, with a total of three biological replicates conducted. Ion leakage of transgenic plant leaves was measured using the methods described previously (Xie *et al.*, 2018). The photosynthesis rate was assessed using a Li-COR-6400 portable photosynthesis system (LI-COR, Nebraska). For water loss rate (WLR), fresh leaves from transgenic and control plants, which were grown for 1 month after soil cultivation, were collected. The leaves were then placed in a weighing dish with the leaf dorsal

surface facing upwards and lost water naturally at room temperature. The fresh weight was measured at specific time intervals, and the water loss rate of the leaves was then calculated as a percentage of the initial fresh weight. Peroxidase (POD) activities, catalase (CAT) activities, and malondialdehyde (MDA) content were measured according to previous methods with minor modifications (Wang *et al.*, 2012). Measurement of relative leaf water content (RWC) was described in a previous study with minor modifications (Li *et al.*, 2019). The H₂O₂ assay kit (#H₂O₂-1-Y, Suzhou Comin Biotechnology Institute, Suzhou, China) was used to determine the hydrogen peroxide (H₂O₂) contents. Leaf H₂O₂ staining was assayed by the same method as previous studies with slight modifications (Guan *et al.*, 2013). The phenotypic assays are described in detail in our previous study (Guo *et al.*, 2023; Li *et al.*, 2023b).

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Conflict of interest

The authors declare that they have no conflict of interest.

Author contributions

Shicong Wang: writing—original draft, validation, visualization, data curation. **Jieqiang He:** methodology, data curation. **Bichun Hu:** validation, data curation. **Mengting Deng:** validation, data curation. **Wenjie Li:** resources. **Junxing Guo:** resources. **Yi Song:** resources. **Qianming Zheng:** methodology, supervision. **Xiaolin Song:** supervision, formal analysis. **Fengwang Ma:** resources, supervision. **Jiangbo Wang:** supervision, funding acquisition. **Qingmei Guan:** conceptualization, investigation, supervision. **Jidi Xu:** conceptualization, writing—review and editing, investigation, supervision, project administration, funding acquisition. All authors have read and agreed to the published version of the manuscript.

Data availability statement

The WGBS, ChIP-seq, and transcriptome data have been deposited into the NCBI BioProject (BioProject, <https://www.ncbi.nlm.nih.gov/bioproject/>) under accession number PRJNA692062.

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Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Figure S1 Characterization of lncRNA and lncRNA regulatory network.

Figure S2 Characterization of DNA methylation and six histone modifications.

Figure S3 Upset plot of differential expression genes co-regulated by DNA methylation and histone modifications of A3d vs A0d (A) and A9d vs A0d (B).

Figure S4 Genomic snapshots of epigenetically regulated genes.

Table S1 Different group GO enrichment.

Table S2 lncRNA GO enrichment.

Table S3 lncRNA miRNA DEGs overlap.

Table S4 lncRNA network.

Table S5 DNAMethy GO enrichment.

Table S6 Histone modification overlap enrichment.

Table S7 Upset plot of genes regulated by different epigenetic marks.